COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9/10/2026	**DATE SUBMITTED:**	9/10/2026
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ACTIVITY NO. 7.1 **Managing Files and Creating Roles in Ansible**

1. PROCEDURE OUTPUT

Task 1: Create a file and copy it to remote servers

1. Using the previous directory that was created in your previous laboratory, create a directory, and name it “files.” Create a file inside that directory and name it “default_site.html.” Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit.

```
momo1@workstation:~/Downloads/CPE212_MOMO$ mkdir -p files
momo1@workstation:~/Downloads/CPE212_MOMO$ nano files/default_site.html
```

momo1@workstation: ~ momo1@workstation: ~/Downloads/CPE212_MO ×

```
 GNU nano 8.7.1                  files/default_site.html *
<!DOCTYPE html>
<html lang="en">
<head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>My Web Server</title>
</head>
<body>

        <h1>Welcome to My Web Server</h1>

        <p>This webpage was deployed using Ansible.</p>
        <p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
```

2. Edit the site.yml file and just below the web_servers play, create a new file to copy the default html file for the site

```
 GNU nano 8.7.1                                site.yml *

- hosts: web_servers
  become: true
  tasks:
    - name: copy default html file for site
      tags:
        - apache
        - apache2
        - httpd
      copy:
        src: files/default_site.html
        dest: /var/www/html/index.html
        owner: root
        group: root
        mode: '0644'
```

3. Run the playbook site.yml. Describe the changes.

```
TASK [copy default html file for site] ***********************************
ok: [192.168.56.103]
ok: [192.168.56.104]
changed: [192.168.56.102]
```

- The 192.168.56.102 or the CentOS managed node changed because Ansible successfully updated the HTML file.

4. Go to the remote servers (web_servers) listed in your inventory. Use cat command to check if the index.html is the same as the local repository file (default_site.html). Do both for Ubuntu and CentOS

- Server 2 (Ubuntu)

```
192.168.56.104 | CHANGED | rc=0 >>
<!DOCTYPE html>
<html lang="en">
<head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>My Web Server</title>
</head>
<body>

        <h1>Welcome to My Web Server</h1>

        <p>This webpage was deployed using Ansible.</p>
        <p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
```

- Server 1 (Ubuntu)

```
192.168.56.103 | CHANGED | rc=0 >>
<!DOCTYPE html>
<html lang="en">
<head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>My Web Server</title>
</head>
<body>

        <h1>Welcome to My Web Server</h1>

        <p>This webpage was deployed using Ansible.</p>
        <p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
```

- Server 3 (CentOS)

```
192.168.56.102 | CHANGED | rc=0 >>
<!DOCTYPE html>
<html lang="en">
<head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>My Web Server</title>
</head>
<body>

        <h1>Welcome to My Web Server</h1>

        <p>This webpage was deployed using Ansible.</p>
        <p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
momo1@workstation:~/Downloads/CPE212_MOMO$
```



5. Sync your local repository with GitHub and describe the changes

```
momo1@workstation:~/Downloads/CPE212_MOMO$ git commit -m "Act 7 Task 1"
[main 8c0b5e4] Act 7 Task 1
 2 files changed, 31 insertions(+)
 create mode 100644 files/default_site.html
momo1@workstation:~/Downloads/CPE212_MOMO$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 791 bytes | 791.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:BorgeMomo/CPE212_MOMO.git
   24a84a7..8c0b5e4  main -> main
momo1@workstation:~/Downloads/CPE212_MOMO$
```

- It was successfully synced

Task 2: Download a file and extract it to a remote server

1. Edit the site.yml. Just before the web_servers play, create a new play

```
- hosts: workstations
  become: true
  tasks:
    - name: install unzip
      package:
        name: unzip

    - name: install terraform
      unarchive:
        src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64>
        dest: /usr/local/bin
        remote_src: yes
        mode: '0755'
        owner: root
        group: root
```

2. Edit the inventory file and add a workstation group. Add any Ubuntu remote server. Make sure to remember the IP address

```
 GNU nano 8.7.1                                inventory.ini *
[web_servers]
192.168.56.103 ansible_user=momo2
192.168.56.104 ansible_user=momo3
192.168.56.102 ansible_user=bmomo

[db_servers]
192.168.56.103 ansible_user=momo2
192.168.56.102 ansible_user=bmomo

[file_servers]
192.168.56.103 ansible_user=momo2

[workstations]
192.168.56.103 ansible_user=momo2
```

3. Run the playbook. Describe the output.

```
PLAY [workstations] ************************************************

TASK [Gathering Facts] *********************************************
ok: [192.168.56.103]

TASK [install unzip] ***********************************************
ok: [192.168.56.103]

TASK [install terraform] *******************************************
ok: [192.168.56.103]
```

```
PLAY RECAP *********************************************************************
192.168.56.102             : ok=10   changed=0    unreachable=0    failed=0    skipped=4
rescued=0    ignored=0
192.168.56.103             : ok=14   changed=1    unreachable=0    failed=0    skipped=5
rescued=0    ignored=0
192.168.56.104             : ok=6    changed=1    unreachable=0    failed=0    skipped=3
rescued=0    ignored=0
```

- We can see that the status for the tasks are “ok”, which means unzip and terraform was already installed.

4. On the Ubuntu remote workstation, type terraform to verify installation of terraform. Describe the output.

```
momo2@Server1:~/Desktop$ terraform
Usage: terraform [-version] [-help] <command> [args]

The available commands for execution are listed below.
The most common, useful commands are shown first, followed by
less common or more advanced commands. If you're just getting
started with Terraform, stick with the common commands. For the
other commands, please read the help and docs before usage.

Common commands:
    apply              Builds or changes infrastructure
    console            Interactive console for Terraform interpolations
    destroy            Destroy Terraform-managed infrastructure
    env                Workspace management
    fmt                Rewrites config files to canonical format
    get                Download and install modules for the configuration
    graph              Create a visual graph of Terraform resources
    import             Import existing infrastructure into Terraform
    init               Initialize a Terraform working directory
    login              Obtain and save credentials for a remote host
    logout             Remove locally-stored credentials for a remote host
    output             Read an output from a state file
    plan               Generate and show an execution plan
```

- We can see that it displayed different commands, which means terraform was successfully installed.

Task 3: Create roles

1. Edit the site.yml. Configure roles as follows: (make sure to create a copy of the old site.yml file because you will be copying the specific plays for all groups)



2. Under the same directory, create a new directory and name it roles. Enter the roles directory and create new directories: base, web_servers, file_servers, db_servers and workstations. For each directory, create a directory and name it tasks.

```
momo1@workstation:~/Downloads/CPE212_MOMO$ ls roles
base  db  db_servers  file_servers  web  web_servers  workstations
```

3. Go to tasks for all directory and create a file. Name it main.yml. In each of the tasks for all directories, copy and paste the code from the old site.yml file. Show all contents of main.yml files for all tasks.

```
momo1@workstation:~/Downloads/CPE212_MOMO$ ls roles/base/tasks
main.yml
momo1@workstation:~/Downloads/CPE212_MOMO$ ls roles/workstations/tasks
main.yml
momo1@workstation:~/Downloads/CPE212_MOMO$ ls roles/web_servers/tasks
main.yml
momo1@workstation:~/Downloads/CPE212_MOMO$ ls roles/db_servers/tasks
main.yml
momo1@workstation:~/Downloads/CPE212_MOMO$ ls roles/file_servers/tasks
main.yml
momo1@workstation:~/Downloads/CPE212_MOMO$
```

```
momo1@workstation:~/Downloads/CPE212_MOMO$ cat roles/base/tasks/main.yml
---
- name: install updates (CentOS)
  tags: always
  dnf:
    update_only: yes
    update_cache: yes
  when: ansible_distribution == "CentOS"

- name: install updates (Ubuntu)
  tags: always
  apt:
    update_cache: yes
  when: ansible_distribution == "Ubuntu"
momo1@workstation:~/Downloads/CPE212_MOMO$ cat roles/workstations/tasks/main.yml
---
- name: install unzip
  package:
    name: unzip

- name: install terraform
  unarchive:
    src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
    dest: /usr/local/bin
    remote_src: yes
    mode: '0755'
    owner: root
    group: root
```

```
momo1@workstation:~/Downloads/CPE212_MOMO$ cat roles/web_servers/tasks/main.yml
---
- name: Install Apache and PHP on Ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: Install Apache and PHP on CentOS
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: Start and enable Apache on CentOS
  service:
    name: httpd
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"

- name: copy default html file for site
  tags:
    - apache
    - apache2
    - httpd
  copy:
    src: files/default_site.html
    dest: /var/www/html/index.html
    owner: root
    group: root
    mode: '0644'
```

```
momo1@workstation:~/Downloads/CPE212_MOMO$ cat roles/file_servers/tasks/main.yml
---
- name: install samba package
  tags: samba
  package:
    name: samba
    state: latest
```

```
momo1@workstation:~/Downloads/CPE212_MOMO$ cat roles/db_servers/tasks/main.yml
---
- name: Install MariaDB on CentOS
  dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: Start and enable MariaDB on CentOS
  service:
    name: mariadb
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"

- name: Install MariaDB on Ubuntu
  apt:
    name: mariadb-server
    state: latest
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: Start and enable MariaDB on Ubuntu
  service:
    name: mariadb
    state: started
    enabled: true
  when: ansible_distribution == "Ubuntu"
```

4. Run the site.yml playbook and describe the output.

```
momo1@workstation:~/Downloads/CPE212_MOMO$ ansible-playbook -i inventory.ini site.yml --ask-become
-pass
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
[WARNING]: Host '192.168.56.104' is using the discovered Python interpreter at '/usr/bin/python3.1
4', but future installation of another Python interpreter could cause a different interpreter to b
e discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_disc
overy.html for more information.
ok: [192.168.56.104]
[WARNING]: Host '192.168.56.103' is using the discovered Python interpreter at '/usr/bin/python3.1
4', but future installation of another Python interpreter could cause a different interpreter to b
e discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_disc
overy.html for more information.
ok: [192.168.56.103]
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.9
', but future installation of another Python interpreter could cause a different interpreter to be
 discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_disco
very.html for more information.
ok: [192.168.56.102]
```

```
TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.103]

TASK [file_servers : install samba package] ************************************************
ok: [192.168.56.103]

PLAY RECAP *********************************************************************************
192.168.56.102             : ok=11   changed=0    unreachable=0    failed=0    skipped=5    rescued=0
ignored=0
192.168.56.103             : ok=15   changed=1    unreachable=0    failed=0    skipped=6    rescued=0
ignored=0
192.168.56.104             : ok=7    changed=1    unreachable=0    failed=0    skipped=4    rescued=0
ignored=0

momo1@workstation:~/Downloads/CPE212_MOMO$
```

- We can see that the playbook ran successfully on all the managed nodes. The tasks were organized into different roles such as base, workstations, web servers, database servers, and file servers. According to the PLAY RECAP, it shows that all servers completed their tasks with zero fail.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of creating roles?
   - Creating roles is important because it can help us organize Ansible tasks into different groups based on their roles and it makes the playbook easier to read and manage. Roles also make automation more organized when managing multiple servers.

2. What is the importance of managing files?
   - Managing files is important because the files need to be properly organized and inserted to the correct servers which makes it easier to copy and manage files automatically without doing it manually on every server which helps save time and keeps the files consistent across different servers.

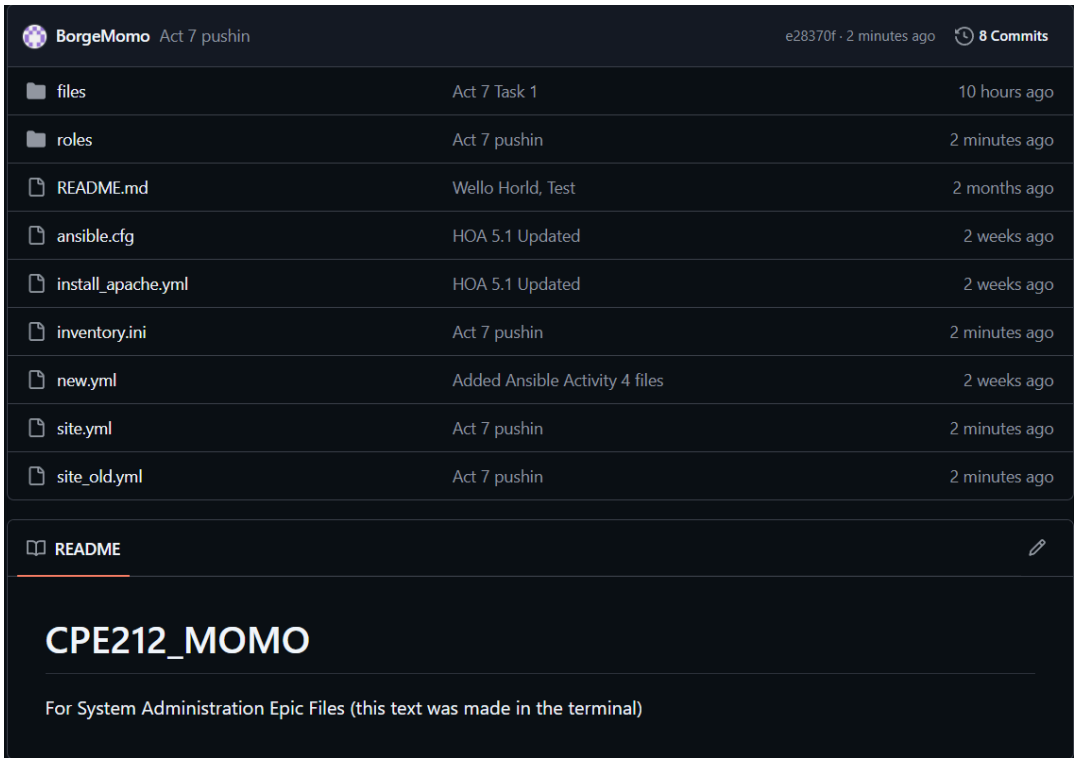
CONCLUSIONS

In conclusion, I learned how to organize Ansible playbooks by using different roles for different tasks and I also learned how roles make the playbook easier to manage and read. Additionally, I noticed that managing files through Ansible is also useful because it allows the files to be copied and inserted to managed servers. Overall, this activity was pretty confusing but along the way, I did all the procedures and understood how Ansible roles and file management can make server automation more organized.

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GitHub Link & Push Proof



https://github.com/BorgeMomo/CPE212_MOMO

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Understanding of Concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **The tool was only used for assistance and to understand some concepts since some parts of the procedure were kind of confusing. The codes and answers were based on my own understanding.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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